

Sustainable Farm Reverse Engineering

2025 Analysis for 2026 Product Transformation

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Product Transformation Lead

Executive Summary

Key Findings

- **2025 Project:** 7-student cohort focused on sustainable mango farming
- **Main Achievement:** Solar tracking system (Arduino-based) with ontology modeling
- **Critical Gap:** Product Transformation was NOT implemented in 2025
- **Industry Reality:** Only 10% has traceability, 15% implements HACCP

2026 Opportunity

Build on 2025 solar tracking work to implement digital batch tracking and HACCP compliance for EU market entry

Sources Analyzed:

- 2025 cohort website (7 participants)
- Yameogo's solar tracking documentation
- Industry research on dried mango processing
- Product Transformation task plan
- Gap analysis matrix

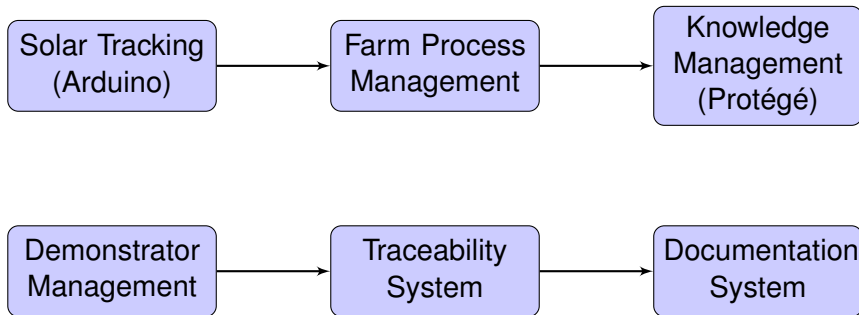
Challenges:

- Binary Word HTML (inaccessible)
- PowerPoint/Excel unreadable
- Week 1 timeline unavailable

Approach:

- Combined docs + industry research
- Focused on Product Transformation
- Identified integration opportunities

2025 System Architecture Overview



- **Key Achievement:** Solar tracking with ontology modeling
- **Missing Component:** Product Transformation batch tracking (new for 2026)

Product Transformation – The 2026 Task

Six Core Areas (New for 2026):

1 **Washing & Sorting**

Yield tracking, first quality gate, water quality

2 **Drying Management**

Core transformation, temp/moisture control, energy tracking

3 **QC Checkpoints**

Quality checks, measurable parameters, pass/fail criteria

4 **Packaging Tracking**

Lot codes, best-before dates, export readiness

5 **Batch Records**

Digital logbook, end-to-end traceability, query capability

6 **Food Safety Compliance**

HACCP implementation, CCP documentation, EU requirements

Data Model for Product Transformation

Core Entity: BATCH (The Traceable Thread)



Process Records

- WASH_SORT_RECORD
- DRYING_RUN
- QC_CHECKPOINT
- PACKAGING_RECORD
- COMPLIANCE_RECORD

Workflow States

RECEIVED → WASHED_SORTED
→ DRYING → DRIED
→ PACKAGED → RELEASED
→ SHIPPED

Industry Standards – Drying Technologies

Technology	Characteristics	Usage
ATESTA Dryer	Gas dryer, 60-65°C, 18-24h	95%
Tunnel Dryer	Modern tunnel ovens, uni-form drying	5%
Solar Dryer	Indirect solar, 20-41°C, 3 days	Research

Key Issues

- Burkina Faso is not a gas producer (economic/ecological threat)
- 100% of units use plastic equipment (accessibility vs. quality)
- Solar opportunity aligns with 2025 tracking work

Critical Processing Parameters

Moisture Content

- Target: 12-18% (ideal)
- Range: 6.12-17.44%
- >18%: Too soft
- <12%: Tough

Drying Time

- Gas: 18-24 hours
- Solar: 3 days
- ATESTA: 22 hours

Drying Temperature

- Gas: 60-65°C
- Solar: 20-41°C
- Oven: 50°C

pH Range

- Acceptable: 3.33-4.94
- Stability: <4.5

Water Supply & Quality

Water Requirements

- Potable water mandatory for food industries
- Source: Wells (60m depth in some cases)
- Storage: Manual extraction to tanks, pipe distribution

Treatment Methods

- Chlorination: Most common
- Hypochlorite Sodium: Recommended for small-scale
- QC: Regular physicochemical/microbiological analysis

Impact: Water quality directly affects first quality gate in Product Transformation

Burkina's Solar Advantage:

- GHI: $\approx 5.5 \text{ kWh/m}^2$
- Harvest period: High radiation
- Favorable for solar drying

Solar Performance:

- Peak: 90°C (collector)
- Chamber: 70°C
- Final moisture: $16 \pm 1.33\%$

Comparison

- **Solar:** 33% yield, 14.17% humidity
- **Oven:** 26% yield, 8.25% humidity

Trade-off:

- Solar takes longer
- Higher yield
- Environmental benefits

Storage & Inventory Management

Storage Conditions

- Temperature: 6-12°C recommended
- Humidity: Compatible with activity
- Duration: 90-day storage tested

Industry Gaps

- **Traceability:** Only 10% implement
- **Batch monitoring:** Not systematic
- **HACCP:** Only 15% implement

Packaging

- Heat-welded polyethylene bags
- Sizes: 1kg and 2kg bags
- Palletized for transport

Market & Export Context

Export Performance

- **Growth:** 602.1 → 4,921.2 tons (1992-2005)
- **Burkina:** Leader in organic dried mango
- **Market Share:** 25% of EU supply

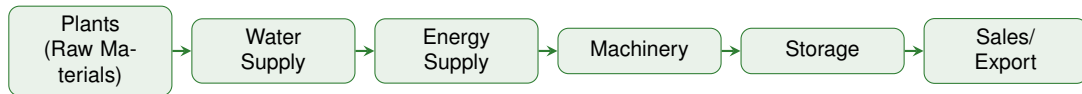
Economic & Ecological Impact

- Thousands of rural women employed
- 1990s Swiss NGO tech still in use
- **Threat:** Imported gas dependency (economic & ecological risk)

Market Requirements

- EU regulations compliance mandatory
- High hygienic standards (HACCP)
- Quality and safety certification

Cross-Functional Dependencies



Critical Dependencies:

- **Water** → **Washing**: Cleanliness directly affects raw intake quality.
- **Energy** → **Drying**: Heating/ventilation control determines final moisture.
- **Machinery** → **Capacity**: Equipment limits control overall processing throughput.
- **Sales/Export** → **Compliance**: EU requirements dictate QC & packaging rules.

Technical Strengths

1 Solar Tracking Innovation

- Arduino-based implementation
- Ontology modeling for integration
- Connection to sustainable farm building

2 Methodological Approach

- Demonstrator method (4-8 week blocks)
- Protégé for ontology development
- Traceability matrices

3 Integration Focus

- Solar energy to farm operations
- Carbon footprint analysis
- Business model templates

2025 Project Gaps

Product Transformation

- No digital batch tracking
- Limited QC documentation
- Food safety (HACCP) not systematically integrated

Documentation Issues

- Binary Word HTML format inaccessible
- PowerPoint files unreadable
- Limited cross-participant visibility

Technical Gaps

- Heavy reliance on gas dryers
- Limited solar drying implementation

Gap Analysis Matrix (2025 → 2026)

Component	2026 Requirement	Priority
Batch Tracking	Digital batch system	HIGH
QC Checkpoints	Formalized at each stage	HIGH
HACCP Compliance	Full implementation	HIGH
Solar Tracking	Integrate with drying	HIGH
Ontology Work	Extend for Product Transformation	MEDIUM
Database	Implement for batch tracking	HIGH
Sensors	Add for process monitoring	HIGH
Mobile Interface	Implement for field data entry	MEDIUM

Summary: 55 gaps identified, 23 marked HIGH priority

2026 Product Transformation Opportunities

1. Digital Batch Tracking

- Database/spreadsheet for batch records
- Lot code generation system
- Batch history query capability

3. HACCP Integration

- Identify Critical Control Points
- Document corrective actions
- Link compliance to batch records

2. QC Checkpoint Automation

- Define measurable parameters
- Establish pass/fail limits
- Create QC specification table

4. Solar Drying Optimization

- Build on 2025 solar tracking work
- Compare carbon footprint
- Integrate solar energy into drying runs

Integration Opportunities

Ontology Extension

- Extend 2025 ontology for Product Transformation entities
- Model BATCH as central entity
- Use Protégé for development

Solar Energy Integration

- Connect solar tracking to drying management
- Track energy source per drying run
- Calculate carbon footprint per batch

Traceability Enhancement

- Expand 2025 traceability matrices
- End-to-end batch traceability

Week 1 Action Plan

Research (Days 1-3)

- Dried mango processing parameters
- QC checkpoints per stage
- HACCP Critical Control Points (CCPs)

Design (Days 4-5)

- Product Transformation data model
- Adapt model with realistic values
- Design QC spec table
- Create HACCP CCP list

Doc (Days 6-7)

- Update task plan with findings
- Create QC spec table
- Document HACCP CCPs
- Prepare summary for homepage

Recommended Next Steps

Product Transformation

- Batch tracking prototype
- QC specification table
- Document HACCP CCPs
- Extend ontology for BATCH

Cross-Collaboration

- Coordinate with Energy Supply
- Work with Water Supply
- Collaborate with Machinery
- Partner with Sales/Export

Cohort Next Steps

- Review gap analysis (55 items)
- Prioritize HIGH gaps (23)
- Assign gap ownership
- Establish weekly tracking

Conclusion

2025 Foundation

- Strong solar tracking and ontology work
- Demonstrator methodology proven
- Documentation structure established

2026 Opportunity

- Address critical dried mango industry gaps
- Build on solar tracking to optimize drying
- Enable EU market entry

Expected Impact

- Digital batch tracking system
- Formalized QC checkpoints
- HACCP compliance documentation
- End-to-end traceability for export



Questions & Discussion

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